Robust Cell Segmentation in Urine Cytology Images for Bladder Cancer Diagnosis

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expensive procedure combined with urine cytology, which is non-invasive but limited by poor sensitivity for detecting low-grade (LG) tumours (Guldhammer et al., 2023; Zhu et al., 2019; Lee et al., 2016). These limitations highlight the pressing need for accurate, non-invasive, and automated diagnostic alternatives. Urine cytology holds potential for early BCa detection, as exfoliated cells from the bladder lining are naturally present in urine (Halachmi et al., 1998). However, brightfield microscopy images used in this context suffer from low contrast, non-uniform illumination, and overlapping cells (Zhu et al., 2019; Gurcan et al., 2009), requiring a set of methods to correctly identify and classify individual cells (Fig. 1).

In urinary cytology, distinguishing between single and clustered urothelial cells is clinically relevant. Single cells are easier to segment due to their clear



Figure 1: Overall segmentation pipeline for urine cytology.

boundaries, whereas clustered cells—common in higher-grade or malignant tumours—appear as overlapping aggregates with ambiguous borders (Owens et al., 2013; Gurcan et al., 2009). The presence of cell clusters exhibiting atypical features is indicative of tumour aggressiveness (Sullivan et al., 2010). Although not all such clusters are malignant, their occurrence in specific architectural patterns (e.g., papillary formations) or in association with distinct cytological atypia may raise suspicion. In low-grade tumours, however, the number of exfoliated single cells and atypical clusters in urine is often limited, posing a

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challenge for accurate cytological detection (Sullivan et al., 2010).

These challenges underscore the need for robust segmentation that can effectively resolve touching or overlapping cells. Traditional methods, like thresholding and watershed, often fail in low-contrast or clustered regions (Gurcan et al., 2009). Deep learning models—U-Net (Zhou et al., 2018), Cellpose (Stringer et al., 2021), and other encoder-decoder architectures—improve accuracy and generalisation across heterogeneous cell types (Ronneberger et al., 2015a; Stringer et al., 2021; Van Valen et al., 2016), but still struggle with overlapping malignant urothelial cells, irregular contours, and high morphological variability (Halachmi et al., 1998; Irshad et al., 2014). Limited annotated datasets and domain-specific models further impede consistent, scalable cytological analysis (Goswami et al., 2021; Litjens et al., 2017).

LG tumours and are confounded by benign conditions, like hematuria (Tomiyama et al., 2024). Newer molecular assays (e.g., UroVysion, CxBladder) and liquid biopsy techniques (Di Meo et al., 2017; Flores Monar et al., 2023) have improved performance but face challenges including biological variability, complex workflows, and limited standardisation (Rashid et al., 2024; Crocetto et al., 2022). Proteomics is another promising avenue but suffers from variability in the urinary proteome and low concordance with tumour tissue (Teixeira-Marques et al., 2023; Wilson et al., 2021).

Advancements in imaging, including fluorescence and confocal cystoscopy, have improved lesion detection (Guldhammer et al., 2023), but require costly equipment and specialised training. In cytology, brightfield microscopy remains the most common approach, but the lack of publicly available datasets has hindered research progress.



2 LITERATURE REVIEW

BCa diagnostics commonly rely on cystoscopy and urinary cytology. While cystoscopy remains the gold standard due to its high specificity, it is invasive, costly, and unsuitable for frequent surveillance. Urinary cytology, though non-invasive, suffers from low sensitivity, particularly for LG tumours, and high inter-observer variability (Lee and Kim, 2020). These limitations have motivated the search for improved non-invasive biomarkers and imaging approaches (Laukhtina et al., 2021).

Urinary biomarkers, such as NMP22 and BTA, offer some promise but show limited sensitivity for

DINO (Caron et al., 2021), which pretrains Vision Transformer (ViT) through self-distillation, enables strong generalisation and can complement Convolutional Neural Networks (CNN)s in low-data regimes. These hybrid approaches combine spatial precision with semantic context, making them suitable for challenging clinical images with morphological variability and limited annotations.

In summary, while classical methods remain relevant, deep learning models—particularly those combining convolutional and transformer features—offer superior adaptability across BCa imaging modalities. These approaches address key challenges such as low contrast, overlapping cells, and domain heterogeneity, positioning them as promising tools for improved cytological diagnostics.

3 METHODOLOGY

The pipeline designed for BCa diagnostics follows a structured sequence of steps that transform raw microscopic images into meaningful diagnostic outputs.
Data Collection. The process begins with the acquisition of brightfield microscopic images of urine samples. Both data quality and quantity are critical for developing robust segmentation systems.
Preprocessing. Raw images often contain noise, illumination inconsistencies, and artefacts. Techniques such as normalisation, contrast enhancement, and noise reduction are applied to standardise the input and improve image clarity.
Cell Segmentation. This step isolates individual cells or cell clusters from the background. Accurate segmentation is fundamental for reliable downstream analysis and for ensuring that only relevant visual in-



sation. Images are originally 3840×2160 RGB TIFFs (4230 total) and are converted to grayscale during preprocessing.

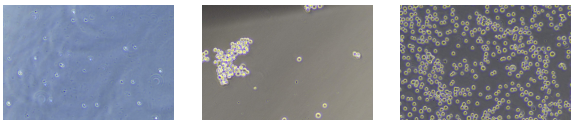


Figure 2: The CELLO dataset's heterogeneity—from isolated to clustered cells.

The dataset's heterogeneity—from isolated to clustered cells—emphasises the need for robust segmentation methods. Additionally, pronounced class imbalance must be addressed to mitigate bias in automated diagnostic pipelines.

3.2 Segmentation

We implemented and evaluated a range of classical and deep learning-based methods for urinary cell segmentation under brightfield microscopy. These techniques address common challenges, including low contrast, uneven illumination, and overlapping cellular structures.
Otsu Thresholding: a global binarisation method that separates foreground and background based on an optimal intensity threshold.
Adaptive Thresholding: local thresholding to handle uneven illumination.
ADAPT: hybrid pipelines combining local thresholding, Gaussian smoothing, and morphological operations.
Cellpose: evaluated in a zero-shot configuration using pretrained *cyto* weights.
DeepCell: our proposed CNN-Transformer teacher

features complement the CNN encoder and are fused via a channel-wise cross-attention layer in the Cellpose decoder. Skip-connected encoder outputs, upsampled and concatenated at matching resolutions, provide both high-level semantic context and fine spatial details, enabling precise cell boundary delineation and robust segmentation in challenging cytology images.

The design leverages Cellpose (Stringer et al., 2021) as a **teacher model** in the DINO configuration to capture firm local texture and morphological cues. In contrast, DINO-finetuned ViT model embeddings (Caron et al., 2021) provide a **student model** that complements the original Cellpose capabilities with extended capabilities to capture long-range contextual information.

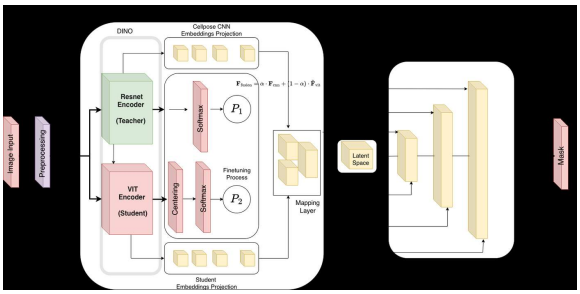


Figure 3: Overview of the DeepCell architecture. DINO features are fused on the encoder output, providing a dense representation for the original Cellpose decoder.

Fusion Strategy. Let $\mathbf{X} \in \mathbb{R}^{H\times W\times 3}$ be an input microscopy image.

- Cellpose encodes $\mathbf{X}$ into CNN model features $\mathbf{F}_{\text{cnn}} \in \mathbb{R}^{H'\times W'\times C_{\text{cnn}}}$, optimized for local cellular boundaries.

Three fusion alternative designs are possible:
Early Fusion: combining CNN and ViT model embeddings immediately after the first feature stage.
Mid Fusion: injecting Transformer embeddings into the Cellpose bottleneck (adopted in our reported results).
Late Fusion: averaging segmentation logits from two independent decoders.

The model was trained using AdamW (weight decay: 1e−4), a learning rate of 1e−3 with cosine decay, a batch size of 16, and for 350 epochs. The loss combined weighted binary cross-entropy, a contour regularisation term, and a teacher–student MSE penalty between CNN features **F**cnn and transformed ViT embeddings $\tilde{\mathbf{F}}$vit ($\lambda = 0.1$). The fused representation is decoded by Cellpose's head into final masks, with skip connections replaced by global attention to propagate ViT-driven context. During initial training, DeepCell's encoders were frozen, and strong augmentations (elastic deformations, blurring, and multiscale sampling) were applied to enhance robustness to overlapping nuclei, irregular boundaries, and variable staining.

3.3 Performance Evaluation

To address variability in image quality and cellular morphology, we developed a multi-stage segmentation pipeline that combines classical image processing with modern deep learning approaches.

Zero-Shot Evaluation. Cellpose serves as a generalist segmentation baseline against Deepcell. We evaluate its pretrained model in a zero-shot setting

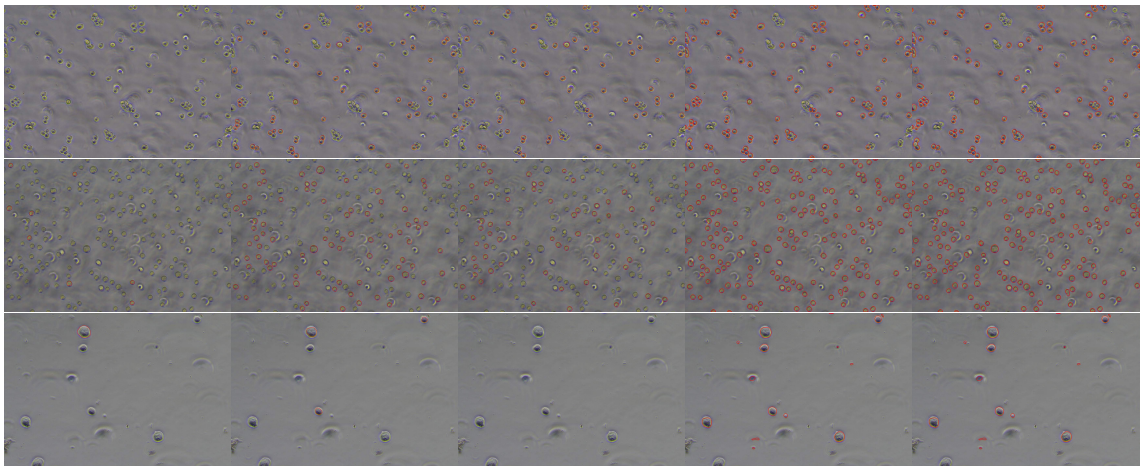


TP (correctly segmented foreground), **FP** (incorrect foreground), **FN** (missed foreground), and **TN** (correct background). From these, we compute three complementary metrics: the **Detection Rate (DR)** for sensitivity, the **False Alarm Rate (FAR)** for oversegmentation, and the **Dice Similarity Coefficient (DSC)** for spatial overlap.

$$\text{DR} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

$$\text{FAR} = \frac{\text{FP}}{\text{FP} + \text{TN}} \tag{4}$$

$$\text{DSC} = \frac{2\cdot\text{TP}}{2\cdot\text{TP} + \text{FP} + \text{FN}} \tag{5}$$



Otsu GAD ADAPT Cellpose DeepCell

Figure 4: Comparison of segmentation methods in the zero-shot CEllo dataset.



Dice score steadily improves during training, reflecting the accurate preservation of contours even in clustered or low-contrast regions.

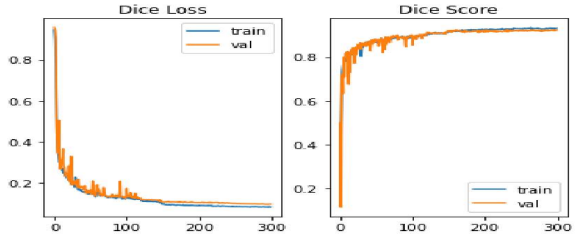
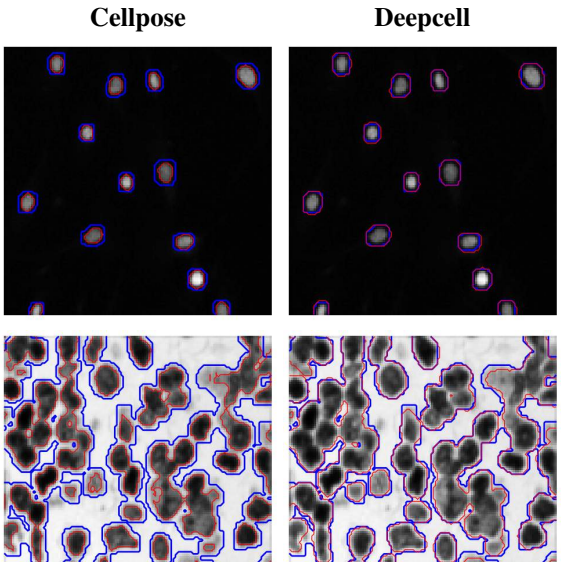


Figure 5: Training DICE loss and score.

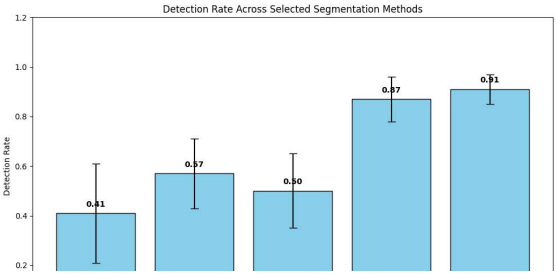
Table 1 summarises the main average methods results on the validation sets regarding the combined BBBC038 (Caicedo et al., 2019), and MoNuSeg (Ku-

DeepCell extends Cellpose by integrating a teacher–student forced learning strategy, which allows it to better capture irregular and clustered cell contours across diverse morphologies and imaging modalities. The model is trained on data spanning varying resolutions, contrast levels, and staining protocols, with extensive geometric, photometric, and resolution-based augmentations during training, further enhancing robustness to scale, illumination, and noise. enhancing its adaptability. By leveraging DINO's self-supervised ViT backbone guided by the teacher's Cellpose encoder, DeepCell embeds both semantic and structural cues, improving boundary delineation in overlapping or low-contrast cells as visible in Fig. 6. As a result, DeepCell surpasses Cellpose in segmenting clustered or morphologically ambigu-



4.3 Zero-Shot Statistical Analysis

To complement the qualitative insights, we performed quantitative assessments by analysing the number of detected contours per image across preprocessing algorithms and patients. Fig. 7 shows that Otsu and GAD detect the most contours on average, though patient-to-patient cell variability—left unnormalized to avoid bias—remains a confounding factor.





in handling irregular or highly variable cell morphologies.

DeepCell and Modified GAD achieve strong zero-shot segmentation across diverse cell morphologies and imaging conditions, effectively separating overlapping or clustered cells and preserving contour accuracy in low-contrast regions. While more computationally demanding, DeepCell offers superior generalisation without extensive parameter tuning. Cellpose baselines demonstrate generalist segmentation capabilities but underperform on domain-specific data with variable staining and morphology. In contrast, DeepCell delivers the most consistent performance across modalities, particularly on the zero-shot Cello dataset, thanks to the combination of self-supervised ViT embeddings and task-specific CNN priors.

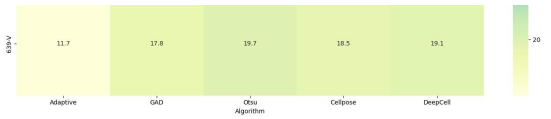


Figure 8: Heatmap showing the number of contours detected per patient and preprocessing method.

Classical methods struggle with irregular, clustered, or atypical cells. DeepCell, a deep learning model fine-tuned for cytology, captures complex and heterogeneous instances more effectively than Cellpose (Stringer et al., 2021), thereby improving segmentation fidelity in high-grade and clustered regions. By combining a teacher–student framework with self-supervised ViT (DINO) features, DeepCell achieves higher detection rates, lower false positives, and im-

proved Dice scores, providing a robust and interpretable toolkit for diverse urine cytology samples.

5 CONCLUSIONS

Our combined qualitative and quantitative evaluations show that segmentation performance is highly dependent on the preprocessing strategy, particularly under variable imaging conditions. Traditional methods, such as Otsu and binary thresholding, perform well in ideal settings but struggle in complex or noisy backgrounds. Adaptive methods offer greater flexibility but may over-segment in cluttered regions. GAD, in particular, yielded the most consistent results for contour continuity and cluster separation. Nonetheless, these approaches still face limitations in handling overlapping, low-contrast, or morphologically



the integration of segmentation into end-to-end diagnostic systems for BCa.

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